HDOC Day-45

Menu-Driven Date Formatting and Substring Generation in C

Problem Statement

Write a menu-driven C program with separate user-defined functions to: (1) convert dd/mm/yyyy into dd-mmm-yy, and (2) print all non-empty contiguous substrings of a string. The program should handle invalid choices and exit when requested.

Algorithm

1. Start and display a menu with Date Formatting, Substrings and Exit options.
2. Read and validate the menu choice.
3. Choice 1: read a date in dd/mm/yyyy format, validate separators, digits and calendar validity including leap years, then display dd-mmm-yy.
4. Choice 2: read a line; for each start position i and end position j, print characters i through j to generate all non-empty contiguous substrings.
5. Choice 3: exit; otherwise display an error and repeat the menu.

Test Case Table

TC	Choice	Input	Expected Output
1	1	12/10/2026	Formatted date: 12-Oct-26
2	1	01/01/2025	Formatted date: 01-Jan-25
3	1	29/02/2024	Formatted date: 29-Feb-24 (leap year)
4	1	29/02/2025	Invalid calendar date.
5	1	31/04/2026	Invalid calendar date.
6	1	12-10-2026	Invalid date format. Use dd/mm/yyyy.
7	2	abc	a, ab, abc, b, bc, c (one per line)
8	2	ab	a, ab, b (one per line)
9	2	(empty input)	Empty string: no non-empty substrings.
10	9	Invalid menu choice	Invalid choice. Enter 1 to 3.
11	3	Exit	Exiting program.

C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
#define MAX 300

void readLine(char text[], size_t size) {
    size_t n;
    int ch;
    if (!fgets(text, (int)size, stdin)) { text[0] = '\0'; return; }
    n = strlen(text);
```

```
    if (n && text[n - 1] == '\n') text[n - 1] = '\0';
    else if (n == size - 1)
        while ((ch = getchar()) != '\n' && ch != EOF) { }
}

int leapYear(int y) {
    return (y % 400 == 0) || (y % 4 == 0 && y % 100 != 0);
}

void changeDateFormat(void) {
    char input[MAX], extra;
    int d, m, y, maxDays;
    const char *months[] = {"Jan", "Feb", "Mar", "Apr", "May", "Jun",
                            "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"};
    printf("Enter date (dd/mm/yyyy): ");
    readLine(input, sizeof input);
    if (strlen(input) != 10 || input[2] != '/' || input[5] != '/' ||
        sscanf(input, "%2d/%2d/%4d%c", &d, &m, &y, &extra) != 3) {
        printf("Invalid date format. Use dd/mm/yyyy.\n");
        return;
    }
    for (int i = 0; i < 10; i++)
        if (i != 2 && i != 5 && !isdigit((unsigned char)input[i])) {
            printf("Invalid date format. Use dd/mm/yyyy.\n");
            return;
        }
    if (y < 1 || m < 1 || m > 12) {
        printf("Invalid calendar date.\n");
        return;
    }
    maxDays = 31;
    if (m == 4 || m == 6 || m == 9 || m == 11) maxDays = 30;
    if (m == 2) maxDays = leapYear(y) ? 29 : 28;
    if (d < 1 || d > maxDays) {
        printf("Invalid calendar date.\n");
        return;
    }
    printf("Formatted date: %02d-%s-%02d\n", d, months[m - 1], y % 100);
}

void printSubstrings(void) {
    char text[MAX];
    size_t i, j, k, n;
    printf("Enter a string: ");
    readLine(text, sizeof text);
    n = strlen(text);
    if (n == 0) {
        printf("Empty string: no non-empty substrings.\n");
        return;
    }
    printf("All substrings:\n");
    for (i = 0; i < n; i++) {
        for (j = i; j < n; j++) {
            for (k = i; k <= j; k++) putchar(text[k]);
            putchar('\n');
        }
    }
}

int main(void) {
    char line[80], extra;
    int choice = 0;
    do {
        printf("\n--- HDOC DAY-45 MENU ---\n");
        printf("1. Change date format\n");
        printf("2. Print all substrings\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        readLine(line, sizeof line);
        if (sscanf(line, " %d %c", &choice, &extra) != 1) {
            printf("Invalid input. Enter 1 to 3.\n");
            choice = 0;
```

```
            continue;
        }
        switch (choice) {
            case 1: changeDateFormat(); break;
            case 2: printSubstrings(); break;
            case 3: printf("Exiting program.\n"); break;
            default: printf("Invalid choice. Enter 1 to 3.\n");
        }
    } while (choice != 3);
    return 0;
}
```

Compilation and Execution Commands

```
gcc hdoc45.c -o hdoc45
```

```
./hdoc45
```

Compilation and Execution



Test Case 1

Choice 1 | Input: 12/10/2026 | Expected: Formatted date: 12-Oct-26



Test Case 2

Choice 1 | Input: 01/01/2025 | Expected: Formatted date: 01-Jan-25



Test Case 3

Choice 1 | Input: 29/02/2024 | Expected: Formatted date: 29-Feb-24 (leap year)

```
Enter your choice: 1
Enter date (dd/mm/yyyy): 29/02/2024
Formatted date: 29-Feb-24
```

Test Case 4

Choice 1 | Input: 29/02/2025 | Expected: Invalid calendar date.

```
Enter your choice: 1
Enter date (dd/mm/yyyy): 29/02/2025
Invalid calendar date.
```

Test Case 5

Choice 1 | Input: 31/04/2026 | Expected: Invalid calendar date.

```
Enter your choice: 1
Enter date (dd/mm/yyyy): 31/04/2026
Invalid calendar date.
```

Test Case 6

Choice 1 | Input: 12-10-2026 | Expected: Invalid date format. Use dd/mm/yyyy.

```
Enter your choice: 1
Enter date (dd/mm/yyyy): 12-10-2026
Invalid date format. Use dd/mm/yyyy.
```

Test Case 7

Choice 2 | Input: abc | Expected: a, ab, abc, b, bc, c (one per line)

```
Enter your choice: 2
Enter a string: abc
All substrings:
a
ab
abc
b
bc
c
```

Test Case 8

Choice 2 | Input: ab | Expected: a, ab, b (one per line)

```
Enter your choice: 2
Enter a string: ab
All substrings:
a
ab
b
```

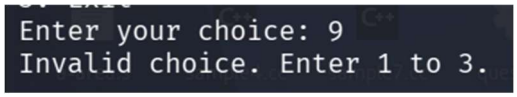
Test Case 9

Choice 2 | Input: (empty input) | Expected: Empty string: no non-empty substrings.

```
Enter your choice: 2
Enter a string:
Empty string: no non-empty substrings.
```

Test Case 10

Choice 9 | Input: Invalid menu choice | Expected: Invalid choice. Enter 1 to 3.



Test Case 11

Choice 3 | Input: Exit | Expected: Exiting program.

